MACHINE LEARNING

ASSIGNMENT - 4

1 C) between -1 and 1

2 D) Ridge Regularisation

3 C) hyperplane

4 A) Logistic Regression

5 B) same as old coefficient of ‘X’

6 C) decreases

7 C) Random Forests are easy to interpret

8 B) Principal Components are calculated using unsupervised learning techniques
C) Principal Components are linear combinations of Linear Variables.

9
A) Identifying developed, developing and under-developed countries on the basis of factors like GDP, poverty index, employment rate, population and living index

B) Identifying loan defaulters in a bank on the basis of previous years’ data of loan accounts.

D) Identifying different segments of disease based on BMI, blood pressure, cholesterol, blood sugar levels.

10
A) max_depth
B) max_features
D) min_samples_leaf

11) **Outliers**

In data analytics, outliers are values within a dataset that vary greatly from the others—they're either much larger, or significantly smaller.

Inter Quartile Range (IQR) –
IQR is used to measure variability by dividing a data set into quartiles. The data is sorted in ascending order and split into 4 equal parts. Q1, Q2, Q3 called first, second and third quartiles are the values which separate the 4 equal parts.

- Q1 represents the 25th percentile of the data.
- Q2 represents the 50th percentile of the data.
- Q3 represents the 75th percentile of the data.

12) Bagging is a method of merging the same type of predictions. Boosting is a method of merging different types of predictions. Bagging decreases variance, not bias, and solves over-fitting issues in a model. Boosting decreases bias, not variance.
In Bagging, each model receives an equal weight. In Boosting, models are weighed based on their performance.
Models are built independently in Bagging. New models are affected by a previously built model's performance in Boosting.
In Bagging, training data subsets are drawn randomly with a replacement for the training dataset. In Boosting, every new subset comprises the elements that were misclassified by previous models.
Bagging is usually applied where the classifier is unstable and has a high variance. Boosting is usually applied where the classifier is stable and simple and has high bias.

13) Adjusted R^2 is a corrected goodness-of-fit (model accuracy) measure for linear models. It identifies the percentage of variance in the target field that is explained by the input or inputs.

R^2 tends to optimistically estimate the fit of the linear regression. It always increases as the number of effects are included in the model. Adjusted R^2 attempts to correct for this overestimation. Adjusted R^2 might decrease if a specific effect does not improve the model.

Adjusted R squared is calculated by dividing the residual mean square error by the total mean square error (which is the sample variance of the target field). The result is then subtracted from 1.

Adjusted R^2 is always less than or equal to R^2. A value of 1 indicates a model that perfectly predicts values in the target field. A value that is less than or equal to 0 indicates a model that has no predictive value. In the real world, adjusted R^2 lies between these values.

14)

S.NO.	Normalization	Standardization
1.	Minimum and maximum value of features are used for scaling	Mean and standard deviation is used for scaling.
2.	It is used when features are of different scales.	It is used when we want to ensure zero mean and unit standard deviation.
3.	Scales values between [0, 1] or [-1, 1].	It is not bounded to a certain range.
4.	It is really affected by outliers.	It is much less affected by outliers.
5.	Scikit-Learn provides a transformer called `MinMaxScaler` for Normalization.	Scikit-Learn provides a transformer called `StandardScaler` for standardization.
6.	This transformation squishes the n-dimensional data into an n-dimensional unit hypercube.	It translates the data to the mean vector of original data to the origin and squishes or expands.

S.NO.	Normalization	Standardization
7.	It is useful when we don’t know about the distribution	It is useful when the feature distribution is Normal or Gaussian.
8.	It is a often called as Scaling Normalization	It is a often called as Z-Score Normalization.

15)

Cross-validation (CV) is a technique used to assess a machine learning model and test its performance (or accuracy). It involves reserving a specific sample of a dataset on which the model isn't trained

Advantage

Cross-validation is used to protect a model from overfitting, especially if the amount of data available is limited.

Disadvantage

The primary challenge of cross-validation is the need for excessive computational resources, especially in methods such as k-fold CV. Since the algorithm has to be rerun from scratch for k times, it requires k times more computation to evaluate.